Rheeya Uppaal

Email: rheeya.uppal@gmail.com | Website: http://uppaal.github.io/

EDUCATION

University of Wisconsin-Madison
Doctor of Philosophy in Computer Science (2022-)

University of Massachusetts, Amherst
Master of Science in Computer Science (2017-19)

Symbiosis Institute of Technology, Pune, India
Bachelor of Technology in Computer Science (2013-17)

PROFESSIONAL AND RESEARCH EXPERIENCE

Amazon Science — New York, June-August 2025
Applied Scientist Intern, under Dr Phu Mon Htut

Amazon Science — Seattle, June-August 2024
Applied Scientist Intern, under Dr Markus Dreyer and Prof Mohit Bansal

UW Madison — Madison, 2022-Present
Research Assistant, *under Prof Junjie Hu*

Goldman Sachs CoreAI — New York City, 2019 – 2021
Research Engineer, *under Dr Vijay Saraswat*

UMass Amherst — Amherst, 2018-2020
Research Assistant, *under Prof Andrew McCallum and Prof Madalina Fiterau*

Goldman Sachs — New York City, June-August 2018
Machine Learning Intern, *under Regina Chan*

Microsoft Research Montreal — Remote, February-July 2018
Research mentee*, under Dr Timothy Hazen*

Indian Institute of Technology Bombay — Mumbai, May-August 2016
Research Intern, *under Prof Virendra Singh*

Indian Institute of Technology Madras — Chennai, January-May 2016
Research Intern, *under Prof Balaraman Ravindran*

PUBLICATIONS

- Ishita Kakkar, Enze Zhang, **Rheeya Uppaal**, Junjie Hu. *When Safety Fails Before the Answer: Benchmarking Harmful Behavior Detection in Reasoning Chains.* Preprint, 2026.
- Xingyu Zhao, Darsh Sharma, **Rheeya Uppaal**, Yiqiao Zhong. *Shattered Compositionality: Counterintuitive Learning Dynamics of Transformers for Arithmetic*. Preprint, 2026.
- **Rheeya Uppaal**, Phu Mon Htut, Min Bai, Nikolaos Pappas, Zheng Qi, Sandesh Swamy. *Journey Before Destination: On the importance of Visual Faithfulness in Slow Thinking.* In the 19th Conference of the European Chapter of the Association for Computational Linguistics (EACL), 2026 (**Oral**). [Article]
- **Rheeya Uppaal**, Apratim Dey, Yiting He, Yiqiao Zhong, Junjie Hu. *Model Editing as a Robust and Denoised variant of DPO: A Case Study on Toxicity.* In the Thirteenth International Conference on Learning Representations (ICLR), 2025. [Talk] [Article]
- Anand Syamkumar, Nora Tseng, Kaycie Barron, Shanglin Yang, Shamya Karumbaiah, **Rheeya Uppaal**, Junjie Hu. *Improving Bilingual Capabilities of Language Models to Support Diverse Linguistic Practices in Education.* Preprint.
- **Rheeya Uppaal**, Yixuan Li, Junjie Hu. *How Useful is Continued Pre-Training for Generative Unsupervised Domain Adaptation?* In the 9th Workshop on Representation Learning for NLP (Rep4NLP) at the 62nd Annual Meeting of the Association for Computational Linguistics (ACL), 2024.
- Yun-Shiuan Chuang, Yi Wu, Dhruv Gupta, **Rheeya Uppaal**, Ananya Kumar, Luhang Sun, Makesh Narsimhan Sreedhar, Sijia Yang, Timothy T. Rogers, Junjie Hu. *Evolving Domain Adaptation of Pretrained Language Models for Text Classification*. In the Workshop on Distribution Shifts: New Frontiers with Foundation Models at the Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS), 2023.
- **Rheeya Uppaal,** Junjie Hu, Yixuan Li. *Is Fine-tuning Needed? Pre-trained Language Models Are Near Perfect for Out-of-Domain Detection*. In the 61st Annual Meeting of the Association for Computational Linguistics (ACL), 2023. [Talk] [Media coverage]
- Ahsaas Bajaj, Pavitra Dangati, Kalpesh Krishna, Pradhiksha Ashok Kumar, **Rheeya Uppaal**, Bradford Windsor, Eliot Brenner, Dominic Dotterrer, Rajarshi Das and Andrew McCallum. *Long Document Summarization in a Low Resource Setting using Pretrained Language Models.* In the Student Workshop at the Joint Conference of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (ACL-IJCNLP), 2021.
- Haw-Shiuan Chang, Shankar Vembu, Sunil Mohan, **Rheeya Uppaal**, Andrew McCallum. *Using Error Decay Prediction to Overcome Practical Issues of Deep Active Learning for Named Entity Recognition.* In European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD), 2020 (Journal Track).
- **Rheeya Uppaal**. *LRS-DAG: Low Resource Supervised Domain Adaptation with Generalization Across Domains.* In the New in ML Workshop of the Thirty-third Conference on Neural Information Processing Systems (NeurIPS), 2019. [Talk]
- **Rheeya Uppaal**, Bryon Kucharski, Bhanu Pratap Singh, Iman Deznabi, Madelina Fiterau. *Multi-resolution Attention with Signal Splitting for Multivariate Time Series Classification.* In the Time Series Workshop of The Thirty-sixth International Conference on Machine Learning (ICML), 2019.

- Bhanu Pratap Singh, Iman Deznabi, **Rheeya Uppaal**, Bryon Kucharski, Bharath Narasimhan, Akhila Josyula, Madelina Fiterau. *Multi-resolution Networks For Flexible Irregular Time Series Modeling*. Preprint.
- Kuiljeit Uppaal, **Rheeya Uppaal**. *Leveraging One's Impact Through Digital Persona and Image Management for Enhanced Effectiveness and Competitive Advantage*. Elsevier, 2018.
- **Rheeya Uppaal**. *Combating Unbalanced Datasets with Generative Adversarial Networks*. Presented at the Women in Data Science Central Massachusetts Conference (WiDS 2018).
- **Rheeya Uppaal**. *Exploring Robust Vehicle Detection and Tracking Methods for Various Illumination Settings using Infrared Thermographic Images*. In the International Journal of Research in Advent Technology, Special Issue National Conference (NCPC 2016).

SERVICE, MENTORING AND OUTREACH

ACADEMIC SERVICE

- Reviewer: COLM (2025-), ICLR (2025-), ACL Rolling Review (2022-), SATA Workshop @ NeurIPS (2024), AKBC (2022), ACM Student Research Competition at the Grace Hopper Conference (2019)
- Member of Department Advisory Board, (2024-)
- Research Team Leadership, Wisconsin AI Safety Initiative (2024-)

INVITED TALKS

- *From Internals to Outcomes: Enhancing Truthfulness and Safety in Language Models*. At Freestone Grove, 2025.
- *Towards Robust Model Editing Techniques.* Cohere for AI's Research Connections Community, 2025.
- *Model Editing for Aligning Models to Preferred Behaviours.* Amazon Science GPS Reading Group, 2024
- *Multi-task Learning in NLP*. UW Madison's CS 769: Advanced Natural Language Processing, Fall 2024.
- *Parameter Efficient Learning*. UW Madison's CS 769: Advanced Natural Language Processing, Fall 2024.
- *Reducing Toxicity in Language Models through Weight Editing.* ML-X Forum at UW Madison, 2024.
- Panellist, *Getting Started in Research.* UW Madison, 2024.
- *Parameter Efficient Learning.* UW Madison's CS 769: Advanced Natural Language Processing, Fall 2023.
- Panellist, *CICS Community Discussion - International Student Experiences*, 2019.

TEACHING AND MENTORING:

- Taught and co-wrote a Carpentries workshop on Trustworthy AI: Validity, Fairness, Explainability, and Uncertainty Assessments, Fall 2024.
- Taught and co-wrote the lecture on Parameter Efficient Fine-tuning in UW Madison's CS 769: Advanced Natural Language Processing, Fall 2023.
- Co-wrote multiple lectures and assignments for UW Madison's CS 220: Introduction to Data Science Programming, Spring 2022.

- Academic Mentor, *ML-X Machine Learning Marathon*, 2023.
- Mentor, *UW Madison ACM-W Year Long Mentorship Program*, 2022, 2023.
- Mentor, *UMass Amherst MS Clusters*, 2021.
- Goldman Sachs Industry Mentor, *UMass Industrial Independent Study*, 2020.
- Mentor and speaker, *UMass CS Women*, 2018-19.
- Mentor, *UMass Women in Engineering Career Day*, 2018.
- Mentor, *Google Summer of Code*, 2016 (representing Systers, an Anita Borg Institute community).

SOCIAL WORK:

- Supporting empowerment of 17 economically challenged girls from India, through sponsorship of their school education till graduation.
- Senator, *UMass Amherst Graduate Student Senate*, 2018.
- Member, *All Ladies League (ALL)*, the world's largest non-profit women's international organization for women empowerment and leadership. Specific focus on supporting economically challenged female students in India.
- Volunteer, *Swacch*, conducting drives across Pune, India to promote the use of paper bags over plastic.
- Volunteer, *Goonj*, participating in drives to collect and collate items for charity across Pune, India.

SELECTED AWARDS AND ACADEMIC ACHIEVEMENTS

- *UW Madison First-year CS Departmental Scholarship*, 2022. (Offered to select students on admission to the university).
- *Karamveer Chakra Award* (instituted by the **United Nations** and iCongo), 2021.
- *Young Leaders Creating a Better World for ALL* Award at Women Economic Forum 2019, for supporting the education of underprivileged school aged girls.
- Semi Finalist, *ACM Student Research Competition –Grace Hopper Celebration for Women in Computing*, 2018.
- All India Rank 2, *National Creativity and Aptitude Test (NCAT)*, 2015.

TECHNICAL SKILLS

Languages: Python, MATLAB, C++, Java, HTML, CSS, JavaScript, MySQL

Deep Learning Frameworks: PyTorch, HuggingFace